

EcoDrive+

Drive Green. Earn Green. Grow Green.

Team Name: Team EcoDrive+

MBOT Fuel Not Found Hackathon 2026

EV Dashboard: Smart Carbon Savings and Eco-Driving Assistant

Proposal Focus	Build Direction
Core idea	A real-time EV sustainability platform that connects ESP32 driving sensors to a polished website dashboard.
Live proof	C++ firmware streams eco-score, harsh events, LED/OLED states, and carbon metrics to the web app.
Creative layer	CarbonTwin forest, EcoCoins, Eco-City builder, community goals, and fleet sustainability reporting.

1. Problem Statement

Electric vehicle drivers often purchase EVs because they want to reduce environmental impact, yet the daily driving experience rarely shows that impact clearly. Current EV dashboards mostly display battery percentage, speed, range, and charging status. These values are useful for operation, but they do not explain how much carbon the driver is saving or how driving habits affect energy consumption.

This creates a behaviour gap. A driver may accelerate too aggressively, brake too late, drive at high drag speeds, or choose routes that waste energy without receiving immediate feedback. The environmental benefit of the EV remains hidden, and hidden impact is difficult to turn into motivation.

The challenge is therefore not only to build a better dashboard. The real challenge is to create a system that makes carbon savings visible, understandable, rewarding, and actionable in real time.

Target Users

- Individual EV drivers who want to understand their real carbon savings and improve range efficiency.
- New EV owners who need simple coaching on smoother acceleration, safer braking, and efficient route choices.
- Student drivers and campus communities who can be motivated through challenges, leaderboards, and shared goals.
- Campus shuttle or institutional EV fleet operators who need sustainability reporting and behaviour insights.

Key user pain points are visibility, feedback, motivation, and proof. EcoDrive+ addresses all four by connecting physical driving behaviour to a live dashboard, carbon metrics, and gamified progress.

2. Solution Description

EcoDrive+ is a real-time EV sustainability platform that combines a live driving dashboard, embedded hardware feedback, carbon impact analytics, and gamified behaviour change. The system uses an ESP32 sensor unit to detect driving behaviour and sends telemetry to a web dashboard where users can see their eco-score, carbon savings, route efficiency, rewards, and community progress.

Module	What It Does	Why It Scores
Live Drive Dashboard	Shows eco-score, speed, distance, kWh estimate, CO2 saved, EcoCoins, and instant driving advice.	Directly solves the dashboard awareness problem.
ESP32 Eco-Feedback Unit	Uses MPU6050, optional GPS, OLED, RGB LEDs, and buzzer to detect and communicate driving behaviour.	Demonstrates real embedded systems and live interaction.
Impact Engine	Converts sensor data into kWh estimate, CO2 saved, range impact, rewards, and route comparison.	Makes sustainability measurable and explainable.
CarbonTwin + Eco-City	Efficient driving grows a forest and earns EcoCoins for an 8x8 sustainable city builder.	Adds creativity, emotion, and long-term engagement.
Community + Fleet	Leaderboards, campus goals, team challenges, and fleet command centre.	Shows wider impact beyond one driver.

The core user loop is simple: drive smoothly, receive immediate feedback, save energy, earn EcoCoins, grow the CarbonTwin forest, build a greener city, and contribute to community carbon goals. This turns sustainability from an abstract number into a visible and repeatable habit.

3. Technical Explanation and Architecture

EcoDrive+ uses both software and embedded systems. The website and backend provide a polished demonstration dashboard, while the ESP32 firmware proves that the idea is connected to real sensor data rather than only a screen simulation.

Layer	Technology	Reason
Firmware	C++ using Arduino framework or PlatformIO	Suitable for ESP32, sensors, OLED, LEDs, buzzer, and the workshop context.
Web App	Next.js, React, TypeScript, Tailwind CSS	Fast to build and polished for a pitching session.
Backend	Node.js, TypeScript, WebSocket server	Reliable real-time telemetry updates during the demo.
Database	SQLite	Local, dependable, and easy to seed without internet dependency.
Visualization	Recharts, Leaflet/OpenStreetMap, optional Three.js	Charts, maps, route comparison, forest, and city visuals.

System architecture:

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ESP32 Sensor Unit -> JSON telemetry over WiFi -> Realtime Gateway -> Impact Engine -> Next.js Dashboard
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The local eco-score uses a weighted formula based on smoothness, braking, speed consistency, and cornering. This transparent method is easy to explain and can run on the ESP32 for low-latency feedback.

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eco_score = smoothness x 0.40 + braking x 0.25 + speed x 0.20 + cornering x 0.15
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A telemetry packet includes the device ID, speed, acceleration, GPS position, eco-score, event type, kWh estimate, and CO2 saved. The website receives it in real time and updates the gauge, event feed, forest, city, leaderboard, and fleet views.

4. Expected Demo: Live Demonstration

The final demonstration uses a physical ESP32 prototype beside a laptop running the EcoDrive+ website. The goal is for judges to see one clear chain: physical movement becomes sensor data, sensor data becomes feedback, and feedback becomes visible carbon impact.

Demo Scene	Action	Expected Result
Problem	Show a normal EV dashboard.	Judges see that battery and range do not explain carbon impact or behaviour.
Smooth Driving	Hold the ESP32 board steadily.	Eco-score stays high, LED is green, OLED shows positive feedback, website shows CO2 saved.
Harsh Event	Jolt or tilt the board.	LED turns red/amber, buzzer beeps, OLED warns, dashboard event feed updates instantly.
Recovery	Return to smooth movement.	Score recovers, advice changes, and EcoCoins are awarded for a smooth streak.
Gamification	Open CarbonTwin and Eco-City screens.	A tree grows and the user places a sustainable building with adjacency bonuses.
Scale	Open community and fleet pages.	Leaderboards, UTAR Green Week, and fleet CO2 savings show wider sustainability impact.

The hardware prototype will use ESP32, MPU6050, optional NEO-6M GPS, SSD1306 OLED, WS2812B LED strip, buzzer, and a push button for demo modes. If GPS signal or WiFi is unstable, the system will include indoor simulator mode and a recorded backup demo. During presentation, we will clearly explain which parts are live and which are simulated.

5. Impact, Future Improvements, and Conclusion

EcoDrive+ creates impact by changing how drivers understand their EV behaviour. Instead of treating sustainability as a hidden number, the system translates each trip into visible carbon savings, energy efficiency, rewards, and progress. This can encourage smoother driving, reduce wasted energy, and make EV ownership more educational and engaging.

For individual drivers, EcoDrive+ provides immediate awareness and positive reinforcement. For students and campus communities, the leaderboard and community goals make sustainability social. For institutions, the fleet mode can support reporting by showing total CO2 saved, average eco-score, harsh braking zones, and vehicle behaviour trends.

The project also has strong educational value. Users learn how acceleration, braking, speed, route choice, and charging decisions affect energy consumption. The Eco-City builder reinforces sustainability concepts by connecting gameplay decisions to real-world ideas such as renewable energy, green infrastructure, and circular economy.

Future Improvements

- Machine learning coaching for personalized eco-driving recommendations.
- More accurate route energy prediction using elevation, traffic, weather, and charging data.
- Real reward partnerships with campus cafes, charging stations, or sustainability programmes.
- Monthly carbon certificates for students, drivers, or fleet operators.
- Expanded fleet analytics for campus shuttle EVs and institutional sustainability reporting.

In conclusion, EcoDrive+ is more than an EV dashboard. It is a complete sustainability feedback loop that senses driving behaviour, explains carbon impact, rewards improvement, and turns individual eco-driving into community progress. By combining ESP32 hardware, C++ firmware, a real-time web dashboard, and gamified sustainability, EcoDrive+ directly addresses the hackathon theme while offering a creative and technically convincing path toward greener driving behaviour.